MLOps
(MACHINE LEARNING OPERATIONS)

→ **MLOps**

It is a set of practices that combines Machine Learning, DevOps & software engineering to Automate, deploy, monitor & maintain machine learning model in production.

→ **Need of MLOps**

MLOps helps to:

- **Automation**: Automate repetitive ML workflows.
- **Reproducibility**: Reproduce experiments using the same data, code & configurations.
- **Deployment**: Make trained models available to users & Applications.
- **Monitoring**: Track model performance & detect problems
- **Versioning**: Manage different versions of datasets, code & Models.
- **Scalability**: Handle increasing workloads & users.
- **Continuous Training**: Retrain models when necessary.

→ ML Development Vs MLOps

Traditional ML	MLOps
(1) Focuses on Model training	(1) Focuses on the complete ML lifecycle
(2) Often involves manual processes	(2) Automates workflows
(3) Model may run locally	(3) Model can run in production
(4) Limited production monitoring	(4) continuous monitoring
(5) Manual model updates	(5) Managed model updates & retraining

→ MLOps Lifecycle

1. **Data collection**
(collect relevant training data)

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2. **Data preperation**
(clean, validate & transform data)

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3. **Model Development**
(Train models & tune parameters)

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4. **Model Evaluation**
(check metrics & acceptance criteria)

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5. **Model Development**
(Integrate the approved model into an application)

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6. **Monitoring**
(Track errors, latency, drift & model quality)

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7. **Feedback & Retraining**
Investigate changes and retrain when necessary

→ DevOps Vs MLOps

	DevOps	MLOps
Main Artifacts	source code, binaries, containers	code, datasets, models, pipelines
Testing	Unit, integration, system tests	software tests plus data & model validation
Versioning	code and releases	code, data, model artifacts, configurations
Monitoring	Availability, errors, latency	Those metrics plus drift & model quality
Updates	software releases	software releases & model updates
Additional challenge	Application behavior	changing data & learned behavior.

→ MLOps Example

House Price Prediction

(1) Train a model using historical housing data.

(2) Track experiments & model versions

(3) Deploy the model in an Application.

(4) Monitor errors, latency & data changes.

(5) If performance drops, retrain, validate & deploy an improved model.

→ Key Terminology

- **ML Pipeline** - A sequence of connected ML processing steps.
- **Model Artifact** - A saved model and its associated files
- **Experiment tracking** - Recording training configurations & results
- **Model registry** - A system for managing model versions & metadata.
- **Model serving** - Making predictions available to Applications
- **Data Drift** - A change in the distribution of input data
- **concept drift** - A change in the relationship bet^{n} input & targets
- **Model decay** - Declining predictive usefulness over time.
- **Rollback** - Returning to a previous working release
- **Retraining** - Training a model again using update data or procedures.